

University of Toronto Mississauga
Mathematical and Computational Sciences
MAT102H5F- Term Test 1
Multiple Choice Booklet
Duration - 110 minutes
No Aids Permitted

This booklet contains the multiple choice questions for **Question 5**. Each part of Question 5 includes five multiple choice answers, of which only a single response is correct. **All answers must be indicated on the scantron sheet (page 6) located at the end of the Test Booklet.** *This* booklet will not be collected at the end of the test, and no work in it will be marked.

You will **not** be submitting this booklet at the end of the test, and are therefore free to remove the staple. The pages have been printed single sided, and are intended to be used for any rough work.

Question 5: Each question is worth 1 mark. Indicate your answer by filling in the bubble sheet on the last page of the exam paper. No part marks will be awarded for this question and you are not required to show your work.

MC 1. Suppose that A, B , and C are sets. Identify the hypothesis of the following theorem:

“If $A \setminus B \subseteq C$ then $A \setminus C \subseteq B$.”

- A. $A \subseteq C$.
- B. $A \setminus C \subseteq B$.
- C. If $A \setminus B \subseteq C$ then $A \setminus C \subseteq B$.
- D. $A \setminus B \subseteq C$.
- E. None of the above.

MC 2. Which of the following describes the set of positive integers that are a power of two? [For this question, you may assume that $0 \in \mathbb{N}$.]

- A. $\{n \in \mathbb{R} : \exists k \in \mathbb{Z}, n = 2^k\}$
- B. $\{n \in \mathbb{N} : \exists k \in \mathbb{N}, n = 2^k\}$
- C. $\{n \in \mathbb{Z} : \exists k \in \mathbb{N}, n = k^2\}$
- D. $\{n \in \mathbb{Z} : \forall k \in \mathbb{Z}, n = 2^k\}$
- E. None of the above.

MC 3. Suppose that $A = \{x \in \mathbb{Z} : \exists y \in \mathbb{Z}, x = y^2\}$ and $B = \{x \in \mathbb{R} : 0 < x^2 < 100\}$. Which of the following is **not** an element of $A \times B$?

- A. $(9, 1)$
- B. $(25, 9)$
- C. $(4, \frac{1}{2})$
- D. $(25, 0)$
- E. None of the above.

MC 4. Suppose that A and B are subsets of $\mathbb{R}$. Which of the following is a correct negation of the sentence:

$$“\forall x \in A, \forall y \in B, \exists z \in A \cap B, |x - y| < 1 \implies |x - z| < 1/2”$$

- A. $\exists x \in A, \exists y \in B, \forall z \in A \cap B, |x - z| \geq 1/2 \implies |x - y| \geq 1$
- B. $\forall x \in A, \forall y \in B, \exists z \in A \cap B, |x - y| < 1 \wedge |x - z| \geq 1/2$
- C. $\exists x \in A, \exists y \in B, \forall z \in A \cap B, |x - y| < 1 \wedge |x - z| \geq 1/2$
- D. $\exists x \in A, \exists y \in B, \forall z \in A \cap B, |x - y| < 1 \vee |x - z| \geq 1/2$
- E. None of the above.

MC 5. Consider the statement “If $m + n$ is odd, then either m or n is odd.” A student gives the following proof:

- (L1) If m is odd then we’re done, so assume that m is even.
- (L2) Since m is even, there is an integer k such that $m = 2k$.
- (L3) Now also assume that n is odd, so that $n = 2\ell + 1$ for some integer ℓ .
- (L4) Then $m + n = (2k) + (2\ell + 1) = 2(k + \ell) + 1 = 2n + 1$, where $n = k + \ell$
- (L5) And this shows that $m + n$ is odd, since we’ve written it as $2n + 1$.

Which line introduces an error into the proof?

- A. L1
- B. L2
- C. L3
- D. L4
- E. The proof is correct.

MC 6. Consider the statement

“Studying hard for a test will result in a good grade.”

Which of the following is a counter-example to this statement?

- A. A test for which you do not study hard and you get a poor grade.
- B. A test for which you do not study hard and get a good grade.
- C. A test for which you study hard and get a poor grade.
- D. A test for which you study hard and get a good grade.
- E. None of the above.

MC 7. Which of the following is a plain English description of the following sentence?

“ $\forall x \in \mathbb{R}, \forall r > 0, \exists q \in \mathbb{Q}, q \in (x - r, x + r).$ ”

- A. There is a rational number in any open interval around a real number.
- B. There is a rational number between any two real numbers.
- C. There is a real number in any open interval around a rational number
- D. For any real number, there is a rational number that is in every open interval around the real number.
- E. None of the above.

MC 8. For which of the following values of P, Q , and R is the statement $(P \wedge Q) \iff (R \wedge \neg R)$ **false**?

- A. P is true, Q is true, and R is true.
- B. P is false, Q is true, and R is false.
- C. P is true, Q is false, and R is true.
- D. P is false, Q is false, and R is false.
- E. None of the above.

MC 9. What is the contrapositive of the following statement:

“Let n be an integer. For every $k \in \mathbb{N}$, if n^k is odd then n is odd.”

- A. For every $k \in \mathbb{N}$, if n is odd then n^k is odd.
- B. For every $k \in \mathbb{N}$, if n is even then n^k is odd.
- C. For every $k \in \mathbb{N}$, if n^k is even then n is even.
- D. For every $k \in \mathbb{N}$, if n is even then n^k is even.
- E. None of the above.

MC 10. What is the contrapositive of the following statement:

“Let x and y be real numbers. If $x < y + \epsilon$ for all $\epsilon > 0$, then $x \leq y$.”

- A. If $x > y$ then for all $\epsilon > 0$ we have $x < y + \epsilon$.
- B. If $x > y$, then there is an $\epsilon > 0$ such that $x \geq y + \epsilon$.
- C. If $x \geq y + \epsilon$ for some $\epsilon > 0$, then $x > y$.
- D. If $x \leq y$, then $x < y + \epsilon$ for all $\epsilon > 0$.
- E. None of the above.